NOMAD Prototype Essay: Voice-Based Cooking Interface

Prototype Innovation and Essay

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INFO 691: Prototyping The User Experience

13 June 2025

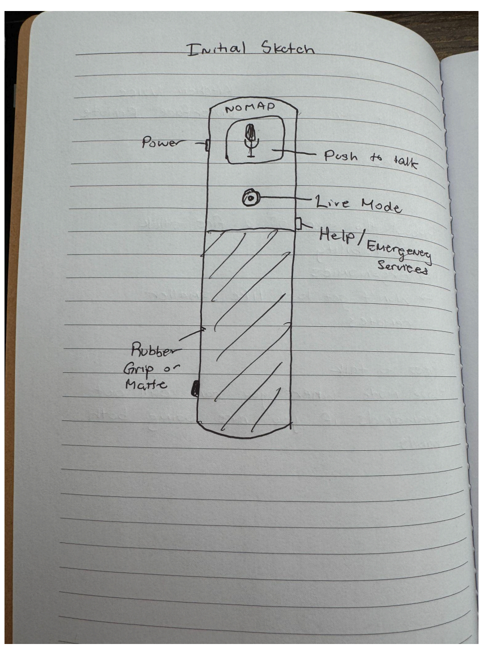
Rationale for the Prototype

For this assignment, I chose to design a voice-based cooking interface called NOMAD (Navigational Oral Module for All Dishes). The goal was to support users in the kitchen by allowing them to control recipe instructions and appliances entirely through voice commands. NOMAD takes the form of a small, ergonomic remote, roughly the size of an Amazon Fire Stick, that users can carry around or place on the counter. It supports both a push-to-talk and a live mode, allowing for seamless interaction whether their hands are free or occupied. I selected this form of the prototype because it reflects realistic constraints and uses contexts, like moving between kitchen tasks or needing to go hands-free while cooking. By creating a physical prototype from existing materials, I was able to focus on tangible interactions and spatial reasoning, key aspects of prototyping embodied interfaces. Additionally, I developed a storyboard and a video demo script to simulate how the product would actually be used. These different modalities helped me convey not only how the user would interact with NOMAD but also how it would behave across various stages of a recipe.

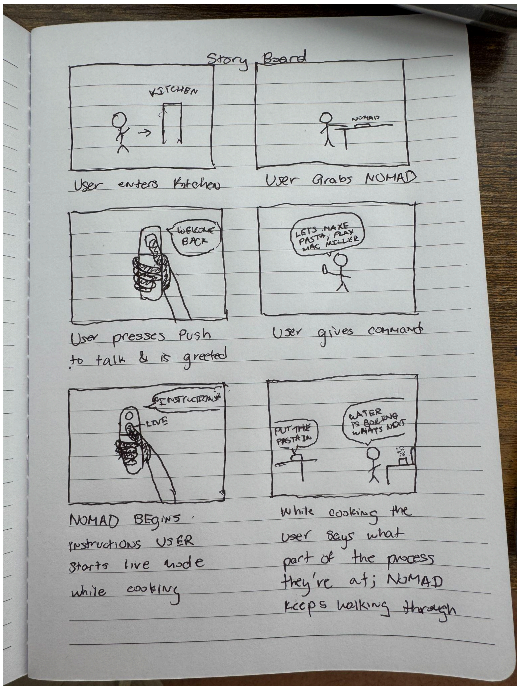
Prototype Choices and Justification

I chose a combination of physical prototyping, storyboarding, and video prototyping because they each highlight different dimensions of the design:

- Physical Prototype: Let me explore the ergonomic form factor and simulate interactions like pressing buttons or placing NOMAD on a surface. This is appropriate for testing affordances and portability.



- Storyboard: Helped visualize the overall use flow—from entering the kitchen to activating NOMAD to interacting while cooking. This storyboard allowed me to plan user scenarios and transitions.



- Video Script: Meant to simulate a full user experience, letting me communicate both functionality and context-of-use through narration and visuals. Seen at the link below:

  NOMAD Demonstration

I deliberately chose not to use high-fidelity UI tools like Figma or InVision, as they are better suited for screen-based apps. For a voice-first, handheld interface, embodied interaction and spatial awareness were more important. Similarly, I avoided Wizard-of-Oz prototyping since the interaction logic was simple enough to demonstrate through scripted voice responses.

These choices align with methods we covered in the course like low-fidelity prototyping for embodied interaction and narrative-based video storytelling as introduced in our readings and video lectures.

Three Memorable Takeaways from This Course

1. **"Prototypes are questions, not answers."**
   This mindset helped me stop obsessing over perfection. Each prototype I created—whether sketch, story, or object—was a chance to ask, *"What would it be like to use this?"*

2. **Fidelity is flexible.**
   I learned that a prototype doesn't need to be fully functional to communicate powerful ideas. Even using Post-its and paper, I could bring NOMAD to life and test core concepts like portability and flow of use.

3. **Prototyping reveals the unexpected.**
   Sketching out the storyboard and building the physical model helped me realize small but important details—like the placement of the emergency button or the need for Live Mode. These weren't in my mind at first but came out naturally during the prototyping process.

Conclusion

All in all, prototyping NOMAD helped me bring a speculative idea into tangible form through thoughtful iteration. By combining physical modeling, storytelling, and scenario simulation, I was able to explore both the functionality and user experience of a voice-based kitchen assistant. This project reinforced the value of low-fidelity prototypes in uncovering design insights early and emphasized how prototyping is not just about presenting solutions, but about asking the right questions. As I continue designing technology for human use, I'll carry forward the mindset that prototypes are a way to think, not just to show.